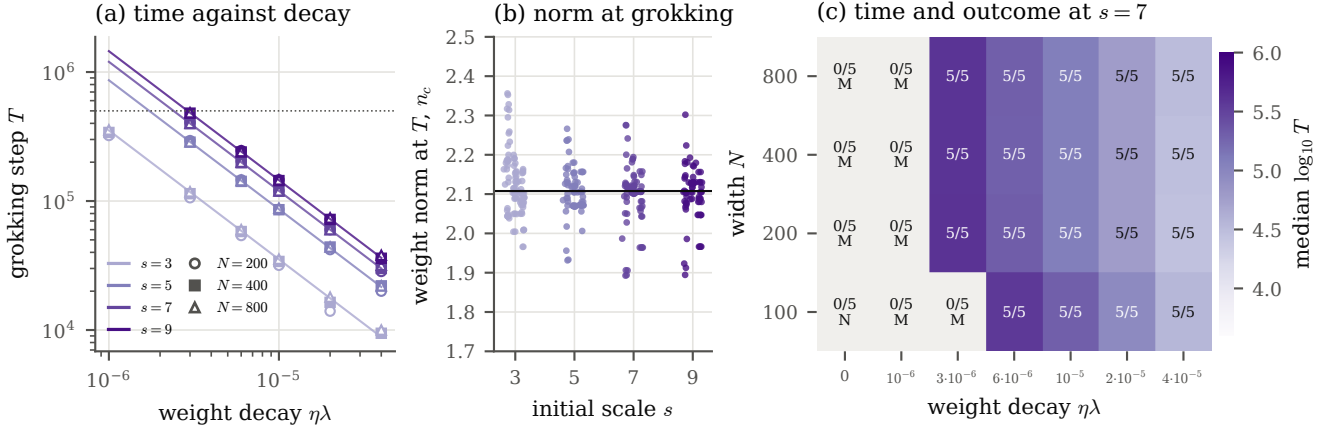
(a) time against decay
grokking step T
10^6
10^5
10^4
10^{-6}
10^{-5}
weight decay $\eta\lambda$
$s = 3$
$s = 5$
$s = 7$
$s = 9$
$N = 200$
$N = 400$
$N = 800$
(b) norm at grokking
weight norm at T, n_c
2.5
2.4
2.3
2.2
2.1
2.0
1.9
1.8
1.7
3
5
7
9
initial scale s
(c) time and outcome at $s = 7$
width N
800
400
200
100
0/5 M
0/5 N
5/5
0
10^{-6}
$3{\cdot}10^{-6}$
$6{\cdot}10^{-6}$
10^{-5}
$2{\cdot}10^{-5}$
$4{\cdot}10^{-5}$
weight decay $\eta\lambda$
median $\log_{10} T$
6.0
5.5
5.0
4.5
4.0